

6. Subqueries

Write and execute subqueries to retrieve data from one table based on results from another

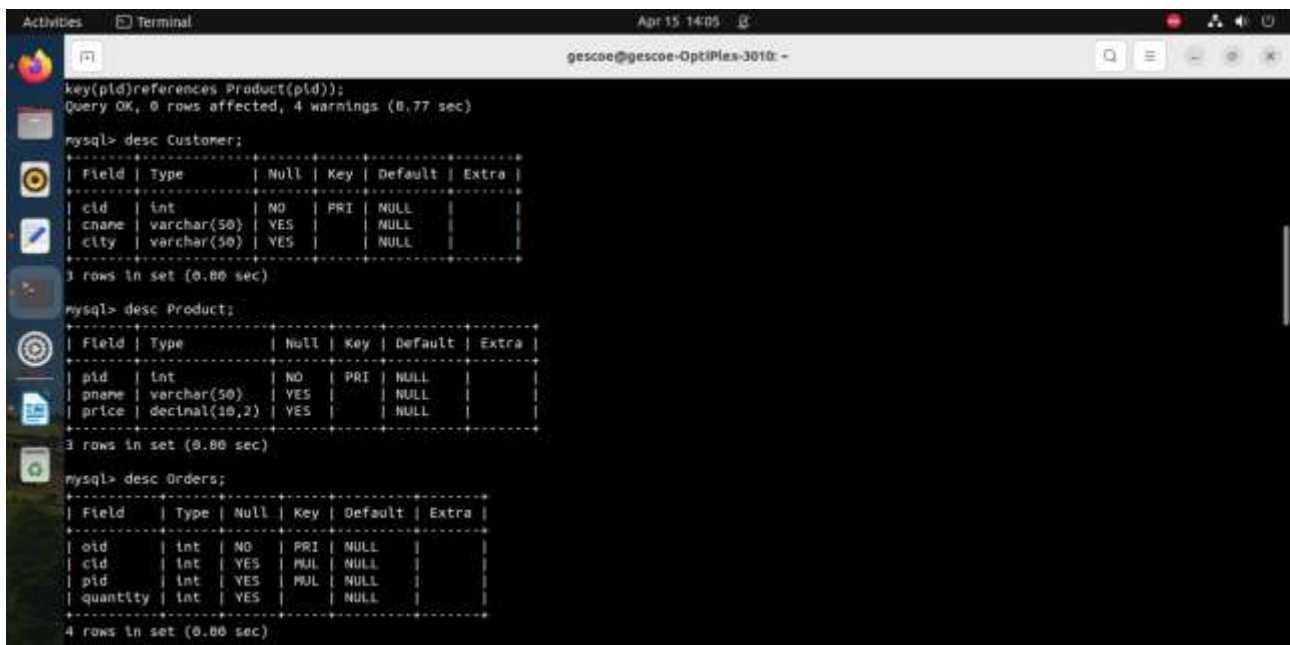
```
mysql> create database Sub;  
Query OK, 1 row affected (0.17 sec)
```

```
mysql> use Sub;  
Database changed
```

```
mysql> create table Customer (cid int (20)Primary key,cname varchar (50),city varchar (50));  
Query OK, 0 rows affected, 1 warning (0.59 sec)
```

```
mysql> create table Product (pid int (20)Primary key,pname varchar (50),price decimal(10,2));  
Query OK, 0 rows affected, 1 warning (0.95 sec)
```

```
mysql> create table Orders (oid int (20) Primary key,cid int(20),pid int(20),quantity int(30),Foreign  
key(cid)references Customer(cid),Foreign key(pid)references Product(pid));  
Query OK, 0 rows affected, 4 warnings (0.77 sec)
```



The screenshot shows a terminal window with the following output:

```
mysql> desc Customer;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| cid   | int  | NO   | PRI | NULL    |       |  
| cname | varchar(50) | YES |     | NULL    |       |  
| city  | varchar(50) | YES |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> desc Product;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| pid   | int  | NO   | PRI | NULL    |       |  
| pname | varchar(50) | YES |     | NULL    |       |  
| price | decimal(10,2) | YES |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> desc Orders;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| oid   | int  | NO   | PRI | NULL    |       |  
| cid   | int  | YES  | MUL | NULL    |       |  
| pid   | int  | YES  | MUL | NULL    |       |  
| quantity | int | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Customer VALUES (1, 'Amit', 'Pune'), (2, 'Neha', 'Mumbai'), (3, 'Rahul',  
'Nagpur');  
Query OK, 3 rows affected (0.13 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Product VALUES (101, 'Laptop', 55000), (102, 'Mobile', 20000), (103,  
'Headphones', 2500);  
Query OK, 3 rows affected (0.11 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Orders VALUES (1001, 1, 101, 1), (1002, 2, 102, 2), (1003, 1, 103, 3);  
Query OK, 3 rows affected (0.10 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
Activities Terminal Apr 15 14:08 gescoe@gescoe-OptiPlex-3010: -
mysql> select * from Customer;
+-----+
| cid | cname | city |
+-----+
| 1 | Amit | Pune |
| 2 | Neha | Mumbai |
| 3 | Rahul | Nagpur |
+-----+
3 rows in set (0.06 sec)

mysql> select * from Product;
+-----+
| pid | pname | price |
+-----+
| 101 | Laptop | 55000.00 |
| 102 | Mobile | 20000.00 |
| 103 | Headphones | 2500.00 |
+-----+
3 rows in set (0.00 sec)

mysql> select * from Orders;
+-----+
| oid | cid | pid | quantity |
+-----+
| 1001 | 1 | 101 | 1 |
| 1002 | 2 | 102 | 2 |
| 1003 | 1 | 103 | 3 |
+-----+
3 rows in set (0.00 sec)
```

Subquery using IN

mysql> Select cname from Customer where cid in (select cid from Orders);

```
+-----+
| cname |
+-----+
| Amit |
| Neha |
+-----+
```

2 rows in set (0.00 sec)

Subquery using NOT IN

mysql> Select cname from Customer where cid not in (select cid from Orders);

```
+-----+
| cname |
+-----+
| Rahul |
+-----+
```

1 row in set (0.02 sec)

Subquery using (Single-Row Subquery)

mysql> SELECT pname, price FROM Product WHERE price = (SELECT MAX(price) FROM Product);

```
+-----+-----+
| pname | price |
+-----+-----+
| Laptop | 55000.00 |
+-----+-----+
```

1 row in set (0.00 sec)

Subquery in SELECT Clause

```
mysql> SELECT oid, (SELECT price FROM Product WHERE Product.pid = Orders.pid) *  
quantity AS total_amount FROM Orders;
```

```
+-----+-----+  
| oid | total_amount |  
+-----+-----+  
| 1001 | 55000.00 |  
| 1002 | 40000.00 |  
| 1003 | 7500.00 |  
+-----+-----+
```

3 rows in set (0.00 sec)

Subquery in FROM Clause

```
mysql> SELECT cid, total FROM (SELECT Orders.cid, SUM(Product.price * Orders.quantity) AS  
total FROM Orders JOIN Product ON Orders.pid = Product.pid GROUP BY Orders.cid) AS  
Purchase WHERE total > 30000;
```

```
+-----+-----+  
| cid | total |  
+-----+-----+  
| 1 | 62500.00 |  
| 2 | 40000.00 |  
+-----+-----+
```

2 rows in set (0.02 sec)

Correlated Subquery

```
mysql> select pname, price from Product P where price > (select Avg(price) from Product where  
Product.pid=P.pid Or 1=1);
```

```
+-----+-----+  
| pname | price |  
+-----+-----+  
| Laptop | 55000.00 |  
+-----+-----+
```

1 row in set (0.00 sec)

Subquery with EXISTS

```
mysql> select cname from Customer C where exists(select 1 from Orders O where O.cid=C.cid);
```

```
+-----+  
| cname |  
+-----+  
| Amit |  
| Neha |  
+-----+
```

2 rows in set (0.00 sec)